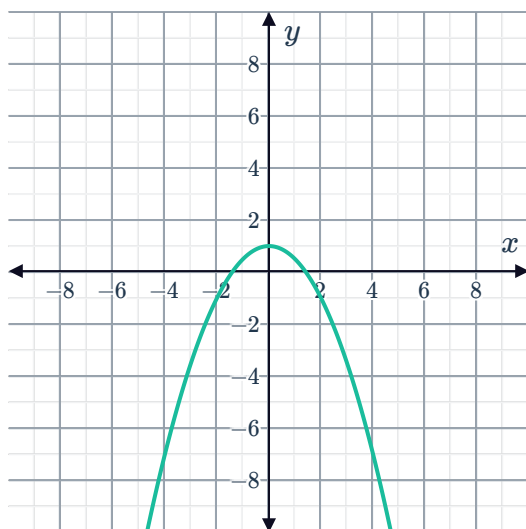


Does an inverse exist?

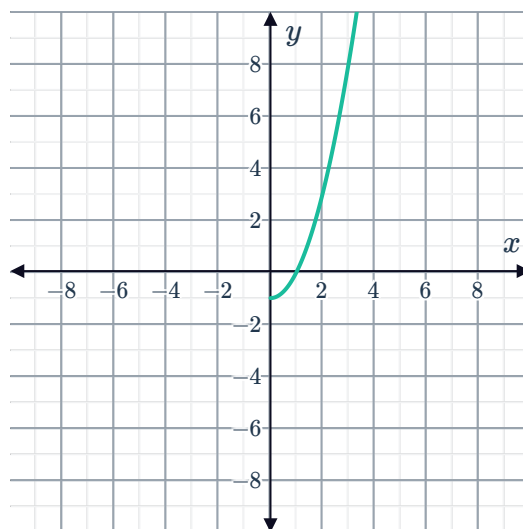


1 For each function drawn below, determine if it has an inverse:

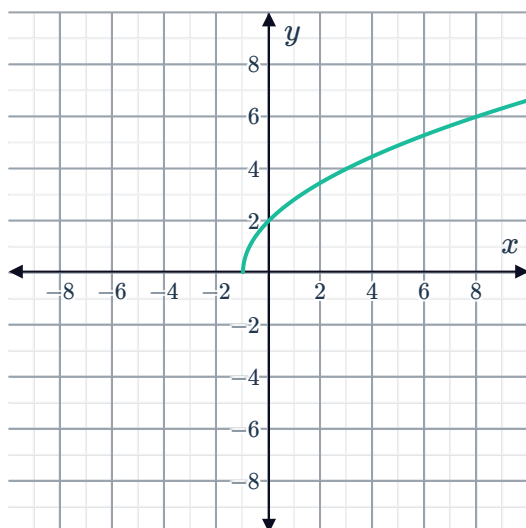
a



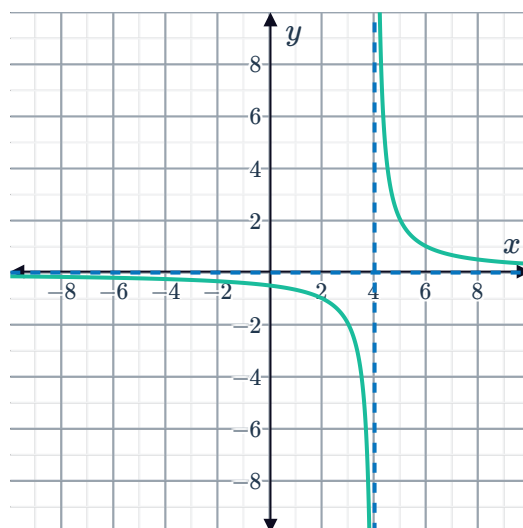
b



c

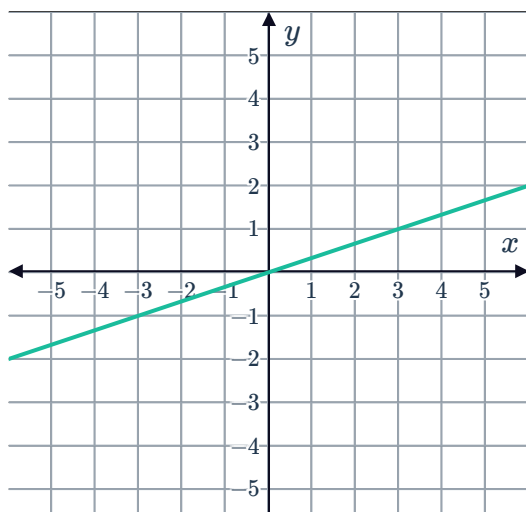


d

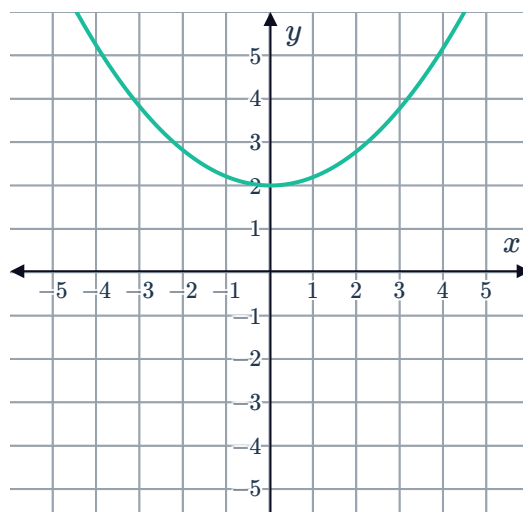


2 State whether the following functions have an inverse function:

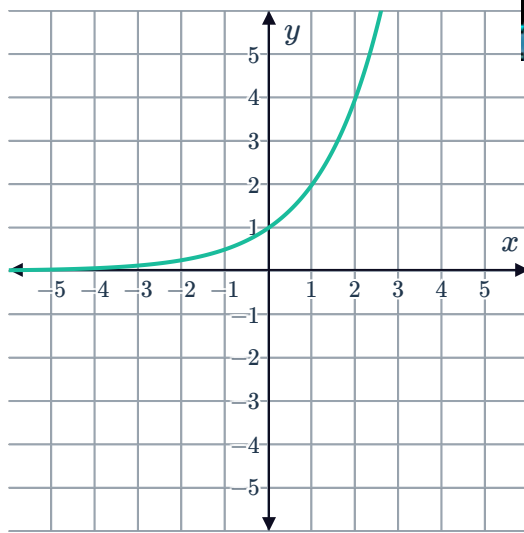
a



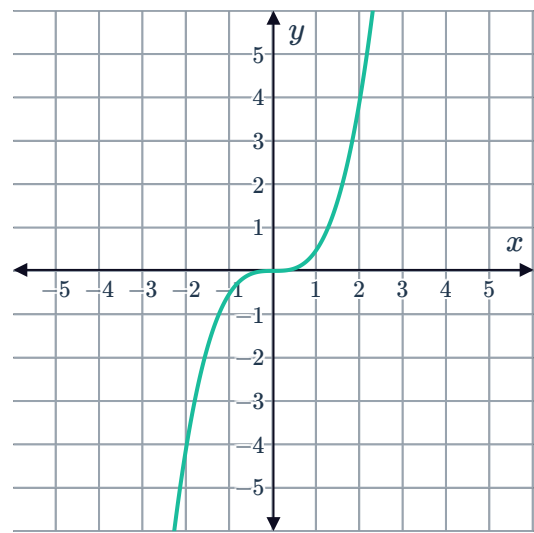
b



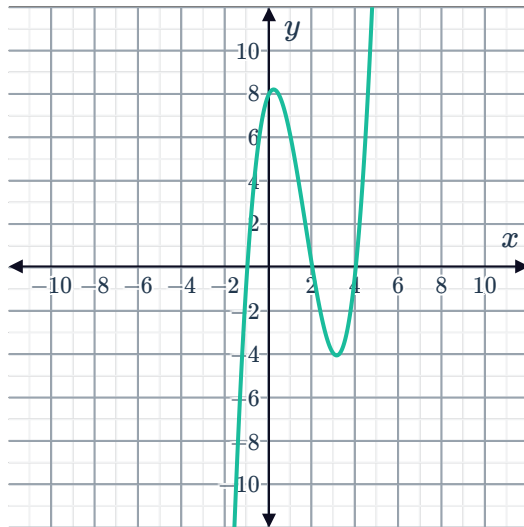
c



d



e



3 Determine whether the following functions have an inverse function:

a $\frac{7}{x}$

b $\sqrt{7-x^2}$

c $\sqrt{x} + 7$

d $1 - 7x$

e 7^x

f $4x^3 + 8$

g $\sqrt{4+x}$

h $-\frac{1}{7}x^2$

4 Consider the function $f(x) = \frac{4}{x+4}$.

a Sketch the graph of the function.

b Is the function a one-to-one function?

c Find an expression for the inverse function $f^{-1}(x)$.

5 For each of the following functions:



- i Sketch the graph of $f(x)$.
- ii State whether the function $f(x)$ is one-to-one.
- iii State whether an inverse function exists.

a $f(x) = x + 5$

b $f(x) = \frac{2x + 3}{2}$

c $f(x) = (x - 2)(x + 3)$

d $f(x) = x^2 - 1$

e $f(x) = (x - 5)^3$

f $f(x) = \frac{1}{8 - x}$ for $x \neq 8$

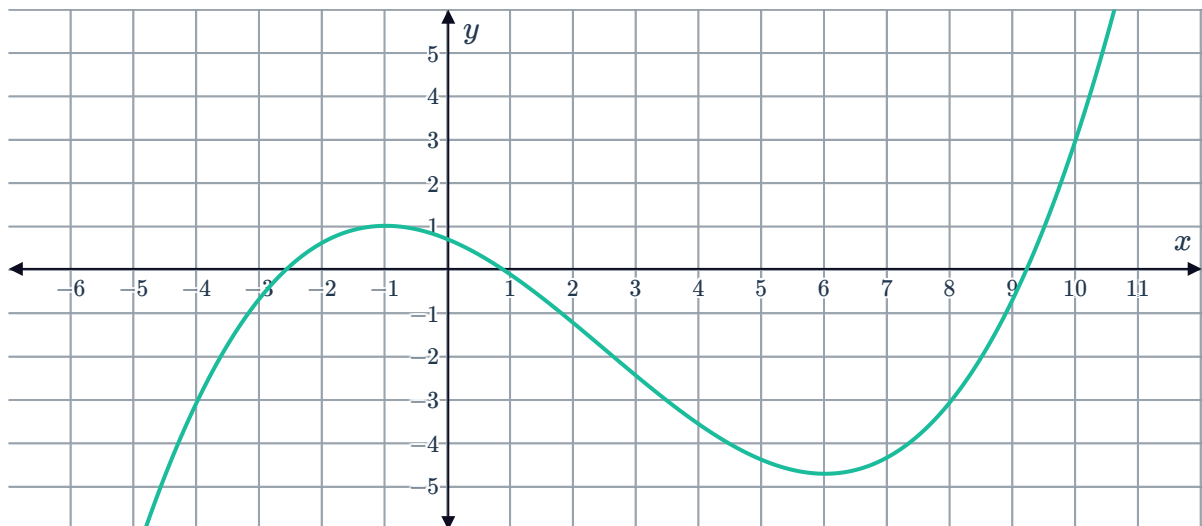
Restricting the domain

6 For many functions, we can determine an inverse relation by first breaking up the original function into parts that are one-to-one, and then finding the inverse of each part separately.

- a Complete the function below to break $f(x) = (x - 2)^2 - 6$ into two one-to-one functions that have the same rule as $f(x)$.

$$f(x) = \begin{cases} (x - 2)^2 - 6, & x < \text{ } \\ (x - 2)^2 - 6, & x \geq \text{ } \end{cases}$$

- b Consider the graph of $y = g(x)$ shown:



Complete the function below to break the graph of $y = g(x)$ into three one-to-one functions that have the same rule as $g(x)$.

$$f(x) = \begin{cases} g(x), & x \leq \text{ } \\ g(x), & \text{ } < x < \text{ } \\ g(x), & x \geq \text{ } \end{cases}$$

- 7 Find an appropriate restricted domain for the function $f(x) = (x - 7)^2 + 12$ to have an inverse.
- 8 Consider the function $f(x) = (x - 4)^2 - 6$ on the restricted domain $[4, \infty)$.
- Find the inverse function $f^{-1}(x)$ for $f(x)$ on the restricted domain.
 - State the domain and range of $f^{-1}(x)$.
- 9 Consider the function $f(x) = (x + 2)^2 - 9$.
- State the domain restriction corresponding to the right half of the parabola described by $f(x)$, such that the restricted parabola is a one-to-one function.
 - Find the inverse function $f^{-1}(x)$ for $f(x)$ on the restricted domain found in part (a).
 - State the domain and range of $f^{-1}(x)$.
- 10 Consider the function $f(x) = (x + 4)^4$.
- State the domain restriction corresponding to the right half of the quartic defined by $f(x)$, such that the restricted quartic is a one-to-one function.
 - Find the inverse function $f^{-1}(x)$ for $f(x)$ on the restricted domain found in part (a).
 - State the domain and range of $f^{-1}(x)$.
- 11 Consider the function $f(x) = -(x - 5)^2$.
- State the domain restriction corresponding to the left half of the parabola defined by $f(x)$, such that the restricted parabola is a one-to-one function.
 - Find the inverse function $f^{-1}(x)$ for $f(x)$ on the restricted domain found in part (a).
 - State the domain and range of $f^{-1}(x)$.
- 12 Consider the function $f(x) = x^2 + 10x + 23$.
- Complete the square for $f(x)$ to write the function in turning point form.
 - State the domain restriction that defines the right half of this function, making it one-to-one.
 - Find the inverse function $f^{-1}(x)$ for $f(x)$ on the restricted domain found in part (b).
 - State the domain and range of $f^{-1}(x)$.

13 Consider the function $f(x) = x^2 + 4x + 3$.



- a Factorise $f(x)$ fully.
- b Find the axis of symmetry of $f(x)$.
- c State the domain restriction that defines the right half of this function, making it one-to-one.
- d Find the inverse function $f^{-1}(x)$ for $f(x)$ on the restricted domain found in part (c).
- e State the domain and range of $f^{-1}(x)$.

14 The inverse of a function $f(x)$ is $f^{-1}(x) = \sqrt{8-x}$.

- a State the domain and range of $f^{-1}(x)$.
- b If $f^{-1}(x)$ is the inverse of $f(x)$, find $f(x)$.
- c State the domain restriction on $f(x)$.
- d What do you notice about the domain and range of $f(x)$ and $f^{-1}(x)$?

15 Consider the function $f(x) = \sqrt{x-5} + 8$.

- a State the domain and range of $f(x)$.
- b Is the function $f(x)$ a one-to-one function?
- c Find the inverse function $f^{-1}(x)$.
- d Given that $f(x)$ is only half a parabola, what must be the domain restriction on $f^{-1}(x)$?
- e What do you notice about the domain and range of $f(x)$ and $f^{-1}(x)$?

16 Consider the function $f(x) = (x-a)^4 + b$.

- a State the largest domain and range of $f(x)$ in terms of a and b that would make $f(x)$ invertible.
- b Find the inverse function $f^{-1}(x)$.
- c State the domain and range of $f^{-1}(x)$ in terms of a and b .

17 Consider the function $f(x) = -(x-a)^2 - a$.

- a What would be the domain restriction on $f(x)$ in terms of a if we only want to keep the right half of the parabola?
- b Find the inverse function $f^{-1}(x)$.
- c Determine the condition on the value of a that results in at least one intersection between the graph of $f(x)$ and the graph of $f^{-1}(x)$.

 18 The largest domain over which the function $f(x) = x^2 + bx + c$ has an inverse is $[3, \infty)$. The domain of the inverse function $f^{-1}(x)$ is $[-2, \infty)$.

a Find the value of b .

b Find the value of c .

Applications

19 An encryption tool takes in a 10-digit code, n , and puts it into the formula $K = (n - a)^4 + b$ to create an encrypted key, K . It is important that when we try to get the code back from the key there is only one possible value for n .

Determine whether the following restrictions should be imposed to ensure the correct code is returned:

a $K > b$

b $n > a$

c $n > b$

d $K > a$

20 The per capita CO₂ emissions of Norway have been recorded since 1950, and the tonnes per capita emissions M , at a time t years after 1950, can be approximated by the model $M = 2.95\sqrt{t} + 7.8$.

a Determine the per capita CO₂ emissions at the time when recording started.

b Express t as a function of M .

c Hence, find the number of years it took for the per capita emissions of 1950 to double. Round your answer to one decimal place.

21 The function $d(t) = 120 - 4.9t^2$ can be used to find the distance d that an object dropped from a height of 120 m has fallen after t seconds.

a Is the function one-to-one?

b Find the inverse function, $t(d)$.

c Prove that $d(t(d)) = d$.

d Prove that $t(d(t)) = t$.

e Are d and t inverse functions?

f Determine how long it will take an object to fall 41.6 m when dropped from a height of 120 m.